DSR301M · PRESENT 2

PBL: COMPARATIVE US AQI TIME SERIES FORECASTING IN CALIFORNIA

WITH EXTREME-EVENT EVALUATION

GROUP 5

OUTLINE

Part 1
Problem & Data Understanding
• Wildfire crisis & historic 496.0 AQI shock
• 3 Critical research gaps (Metric, Leakage, Weather)
• 181k Data audit & Jensen's inequality proof

Part 2
Feature Engineering & Visualization
• Zero-leakage temporal split (Train/Val/Test)
• Trim tail buffer defense against label shift
• Physics-informed feature store (VPD, wind u, v)

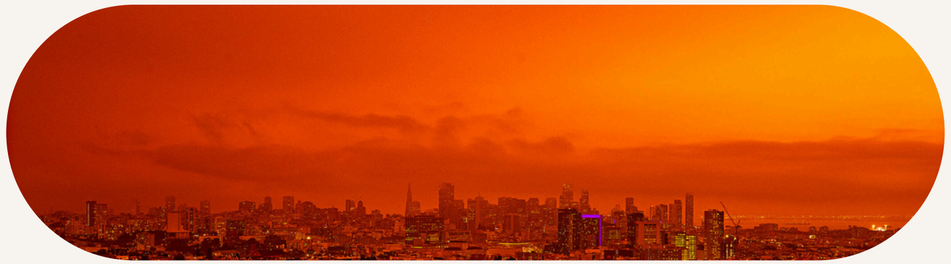
Part 3
Modeling, Evaluation
• Official Test 2025 leaderboard & 168h tracking
• Wildfire shock resilience, TreeSHAP & 3D diagnostics

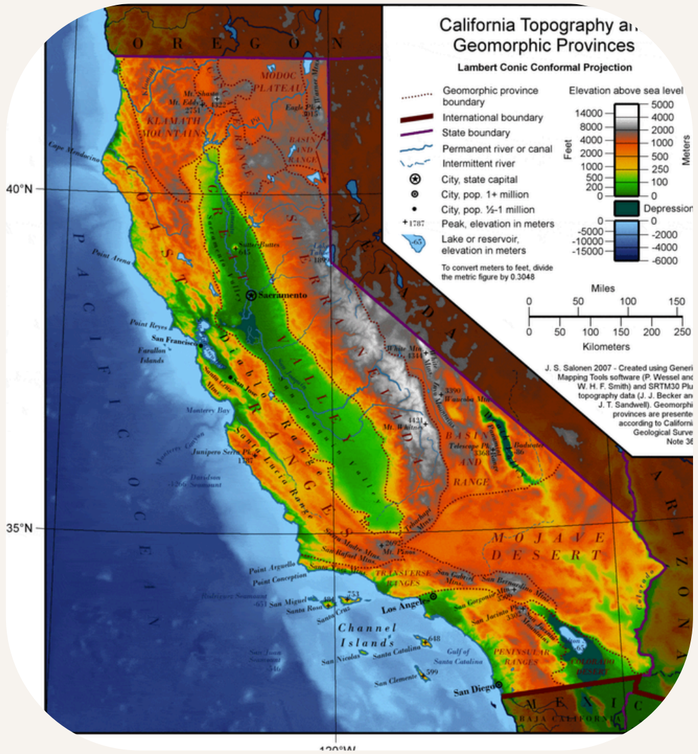
Part 4
Conclusion
• 4 Empirical breakthroughs & 3 technical limits
• Real-time California air quality dashboard & Q&A

PART 1: PROBLEM & DATA UNDERSTANDING

Wildfire Crisis & Hazardous Smoke Hazards in California

Historic August Complex 2020 Megafire & Acute Respiratory Shocks





HISTORIC AUGUST COMPLEX PEAK: 496.0 AQI

- Surpasses Hazardous threshold (301+) by +195 pts
- San Jose (Sept 2020) · Daylight blotted out across Bay Area

ACUTE PUBLIC HEALTH BURDEN

- >90% surge in emergency hospital visits
- Wildfire PM2.5 exceeded WHO guidelines by >10x

SEVERE DATA IMBALANCE

- >85% of days are clean air baseline (AQI ≤ 50)
- Standard models miss rare disaster spikes

PART 1: PROBLEM & DATA UNDERSTANDING

Three Research Gaps & Dual Operational Horizons

Designing Purpose-Built Forecasting Architectures for Public Health Action

GAP 1: METRIC MASKING

- Clean days (>85%) dominate average MAE
- Masks catastrophic wildfire errors (AQI > 150)

GAP 2: LOOKAHEAD LEAKAGE

- Random split leaks future time steps
- Illusory $R^2 > 0.95$ collapses in live production

GAP 3: ATMOSPHERIC BLINDSPOT

- Sensor autocorrelation vanishes by 24h
- Blind without Numerical Weather Prediction (NWP)

HORIZON 1 · NOWCASTING 1h (t+1)

- Action: Immediate worker alerts & traffic routing
- Driver: Sensor inertia (lags 0–3h, rolling stats)

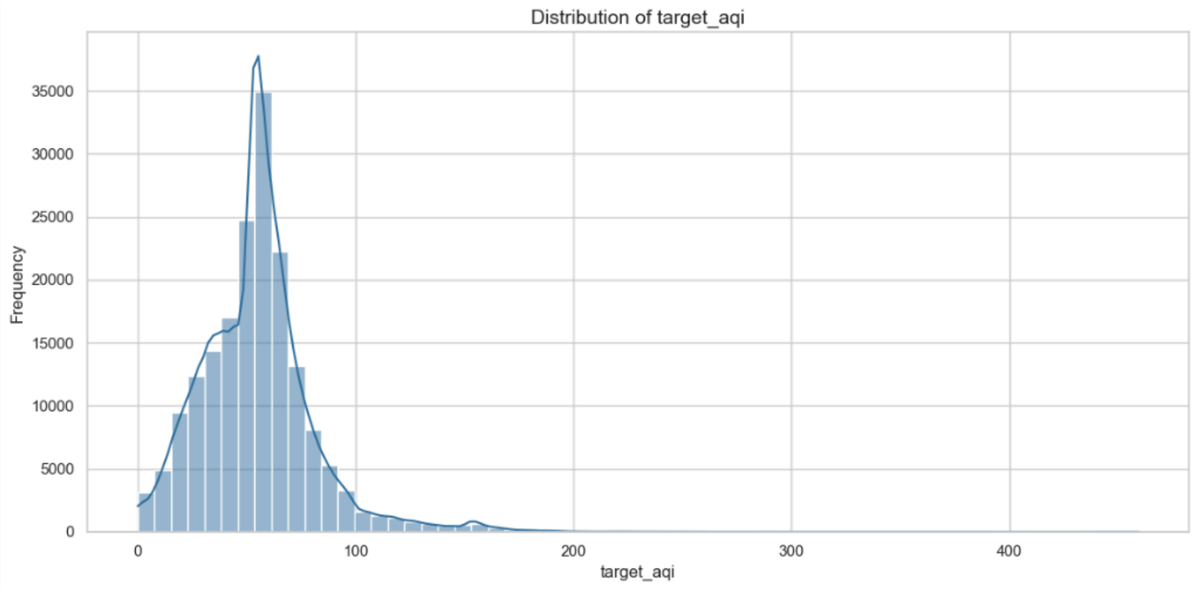
HORIZON 2 · DAY-AHEAD 24h (t+24)

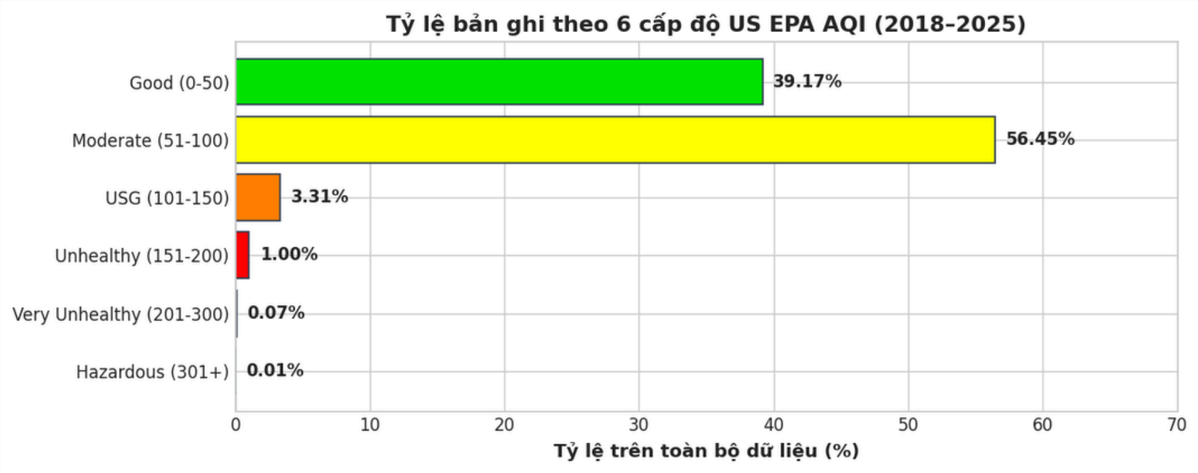
- Action: School closures & ICU hospital staging
- Driver: Zero future sensors; powered by NWP weather

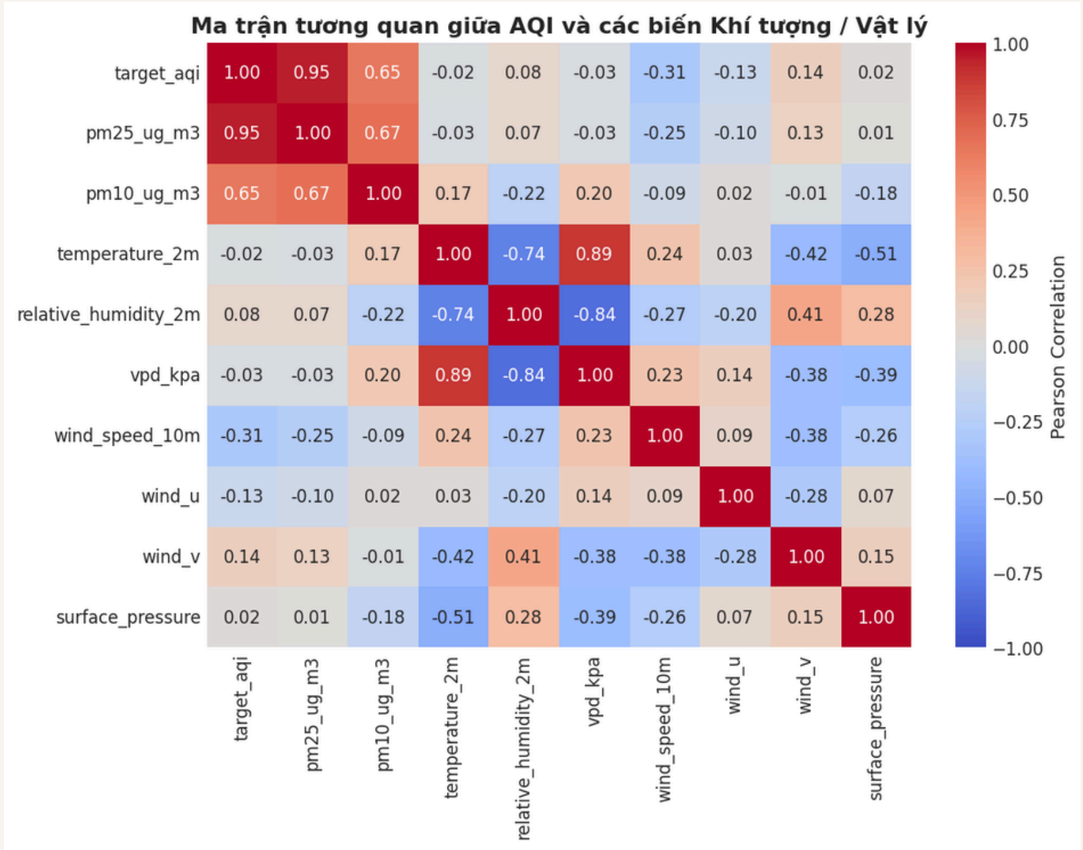
PART 1: PROBLEM & DATA UNDERSTANDING

181k Dataset Audit & Mathematical Distribution Defense

181,122 Hourly Records (2018–2025) Across 3 Topographic Basins



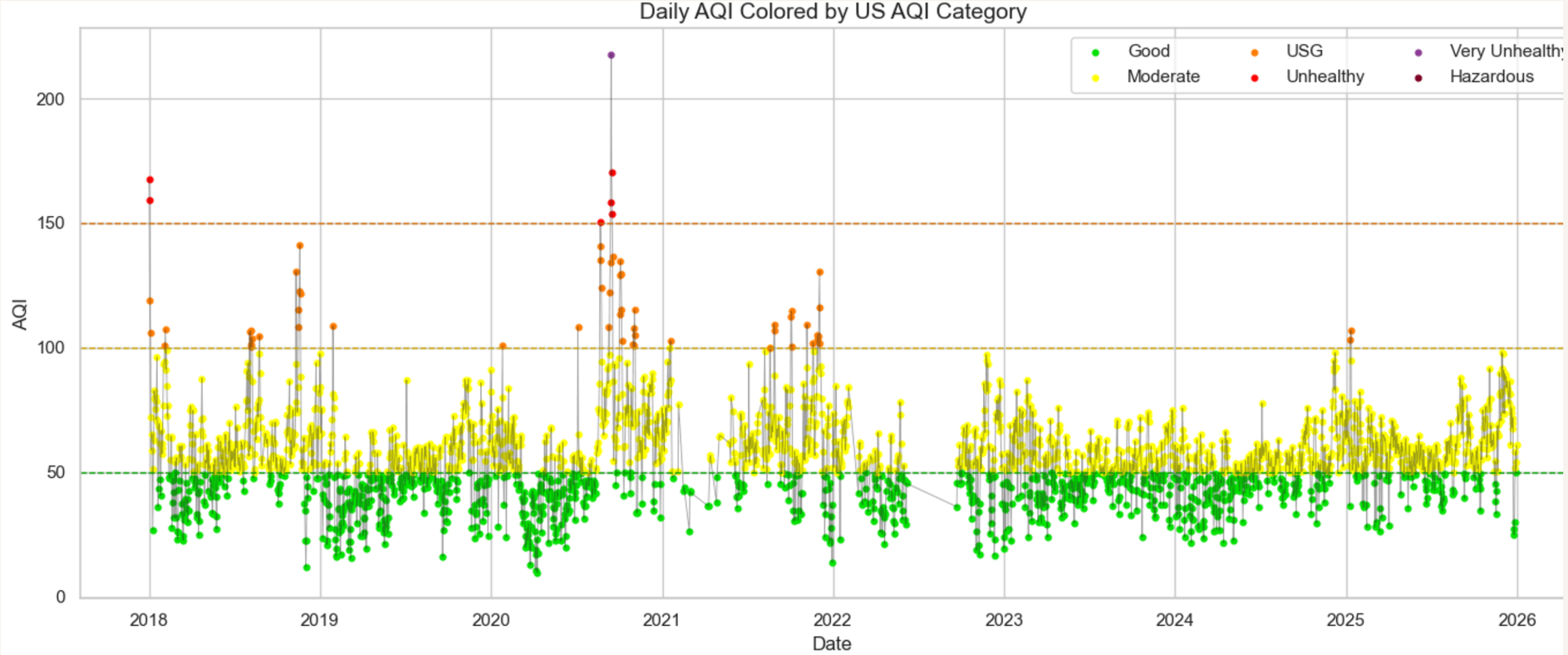




PART 1: PROBLEM & DATA UNDERSTANDING

Zero-Leakage Temporal Split Protocol & Buffer Defense

8-Year Sequential Partitioning Preserving Historical Independence



TRAIN SET · 2018–2023 (61.3% | 111,073 rows)

VALIDATION SET · 2024 (24.1% | 43,727 rows)

TEST SET · 2025 (14.5% | 26,292 rows)

PART 2: FEATURE ENGINEERING & VISUALIZATION

Physics-Informed Feature Store Architecture

Specialized Multi-Horizon Feature Spaces Coupling Surface Sensors with Mesoscale NWP

NOWCASTING 1h (22 Features)

- Autoregression: lag_0..3 + rolling mean/std (3h)
- Plume Momentum: diff_1h (AQI_t - AQI_t-1)
- Surface Weather: Temp, humidity, rain, pressure
- Wind Vectors: Orthogonal components (u, v)
- Calendar Cycles: Cyclical hour & month (sin/cos)

DAY-AHEAD 24h (19 Features)

- Sensor Isolation: Zero future sensor access (t+1..24)
- Base History: lag_0, lag_24, lag_48, roll_mean_24h
- Future Weather: 24h forecast temp, RH, rain, VPD
- Future Wind: Forecast vectors (u, v) & speed
- Future Calendar: Forecast hour & month (sin/cos)

ATMOSPHERIC PHYSICAL DRIVERS

- **Wind Vectors (u, v): Eliminates 359° ↔ 1° polar angle discontinuity**
- **Vapor Pressure Deficit (VPD): Quantifies wildfire fuel aridity & spread**

PART 3: MODELING, EVALUATION

Benchmark Model Suite: 5 Algorithm Families

Standardized Modeling Pipeline from Analytical Baselines to Tabular Deep Learning

BASELINES
- Persistence (y_t)
- Climatology (Hist. mean)
- Zero-training benchmark

LINEAR L2
- Ridge Regression
- Analytical closed-form
- Fit time: ~0.18s

BAGGING
- Random Forest
- 150 Decision Trees
- Subsampling variance cut

GBDT SOTA
- LightGBM (leaf-wise)
- XGBoost (exact Hessian)
- CatBoost (oblivious)

DEEP LEARNING
- Tabular MLP Regressor
- 2 Layers (64, 32 units)
- ReLU, Adam, Early stop

TOP-3 GBDT VOTING ENSEMBLE: Unweighted blending of LightGBM + XGBoost + CatBoost to eliminate idiosyncratic error.

DIFFERENTIAL FEATURE SCALING
- Tree Models (GBDT, RF): Raw unscaled data (scale-invariant)
- Ridge & MLP: StandardScaler fit strictly on Train only (zero leakage)

SPATIAL CATEGORICAL ENCODING
- Ridge & MLP: One-Hot Encoding (prevents false numeric distance)
- GBDT Suite: Native Categorical (optimal histogram split by station)

PART 3: MODELING, EVALUATION

Official Leaderboard: Test 2025 Out-of-Sample Evaluation

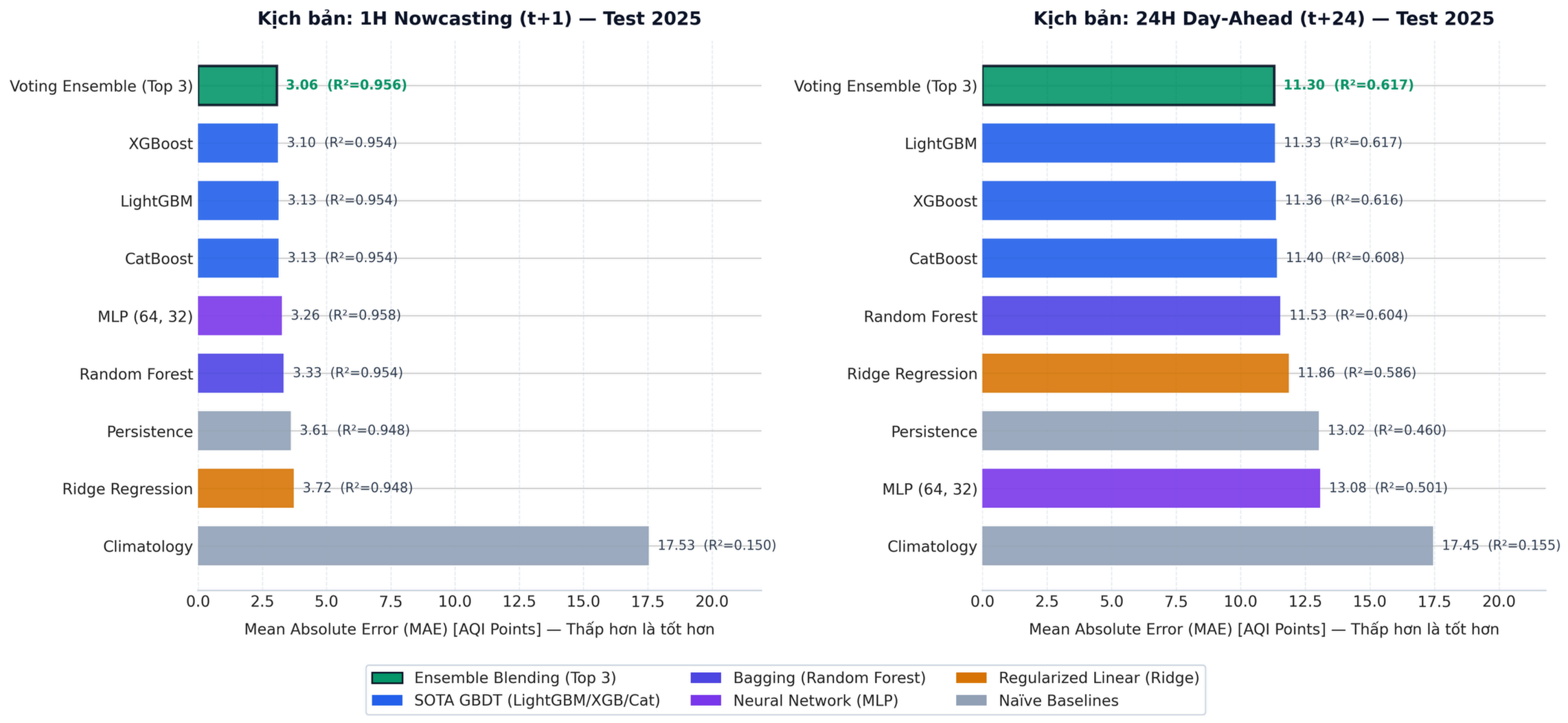
Comparative Performance on 26,200 Unseen Points (comprehensive_leaderboard.csv)

Model	MAE 1h	R^2 1h	MAE 24h	R^2 24h	Fit (24h)
Climatology	17.53	0.150	17.45	0.155	1.44s
Persistence	3.61	0.948	13.02	0.460	0.00s
Ridge Reg.	3.72	0.948	11.86	0.586	0.18s
LightGBM	3.13	0.954	11.33	0.617	0.41s
XGBoost	3.10	0.954	11.36	0.616	0.67s
CatBoost	3.13	0.954	11.40	0.608	1.45s
Tabular MLP	3.26	0.958	13.08	0.501	81.62s
Voting Ensem.	3.06	0.956	11.30	0.617	0.20s

PART 3: MODELING, EVALUATION

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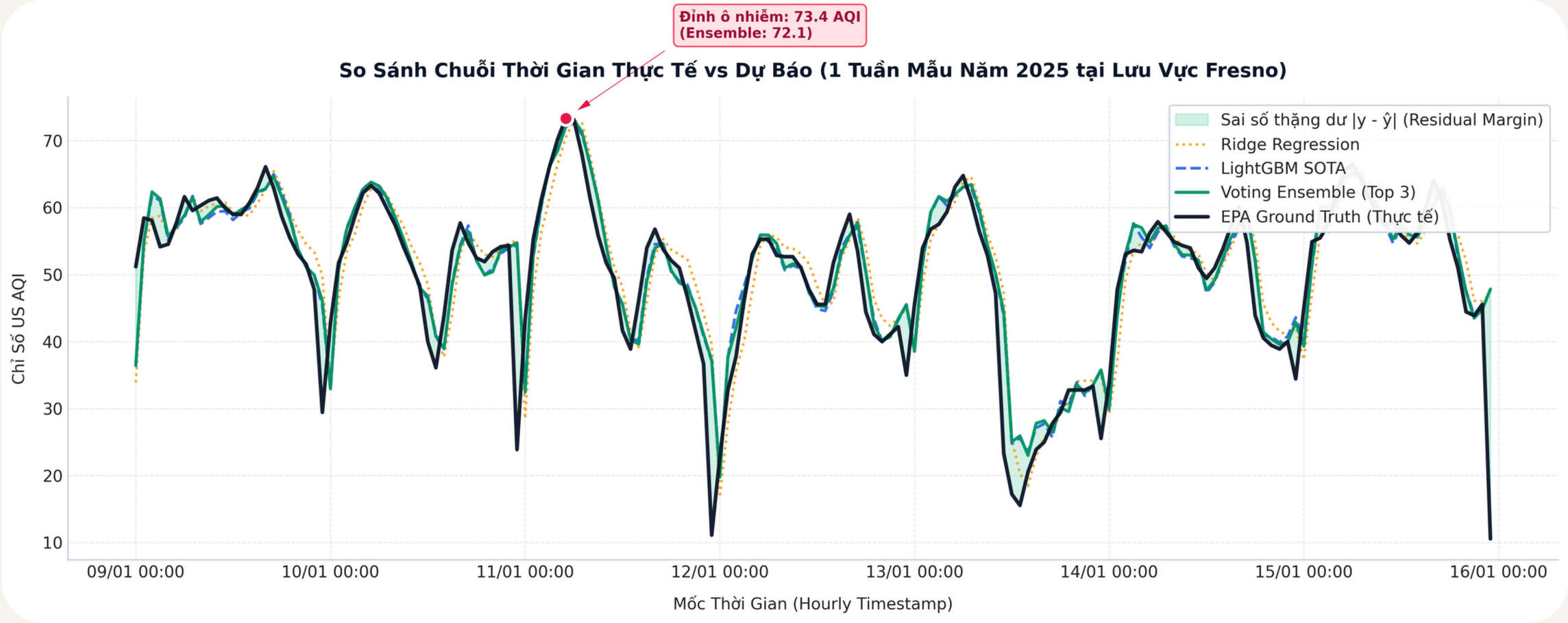
Comparative Performance on 26,200 Unseen Points (comprehensive_leaderboard.csv)



PART 3: MODELING, EVALUATION

168-Hour Continuous Tracking & Residual Dynamics

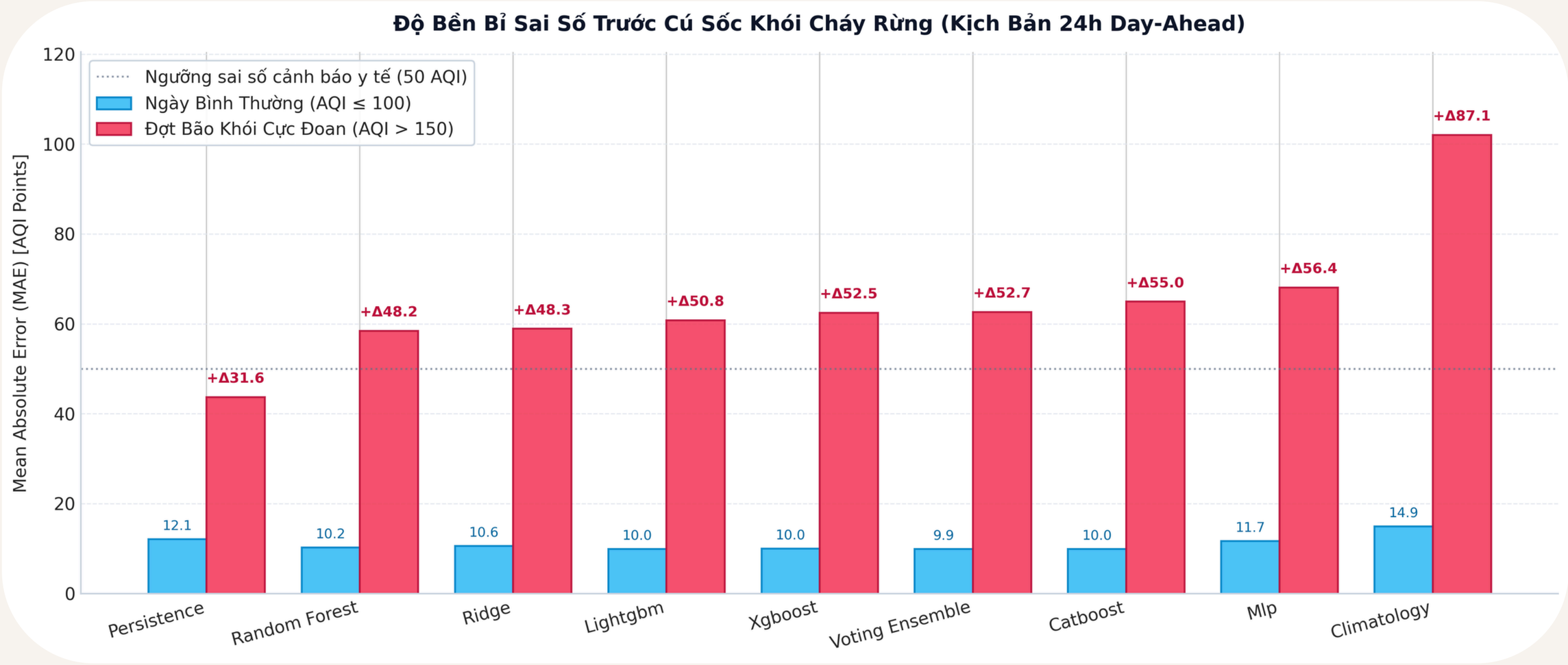
January 09–15, 2025 at Fresno Central Valley Inversion Basin



PART 3: MODELING, EVALUATION

Wildfire Extreme Shock Resilience (24h Day-Ahead)

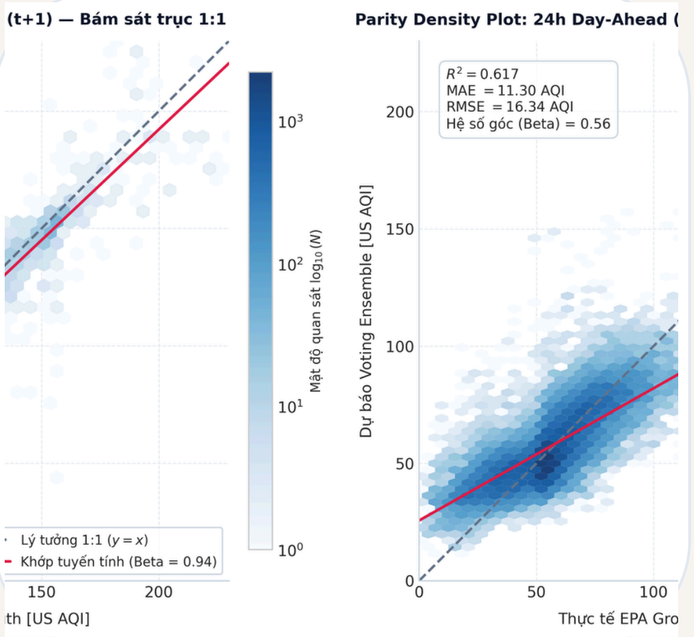
Quantifying the Extreme Shock Gap and Resolving Variance Restriction Artifacts

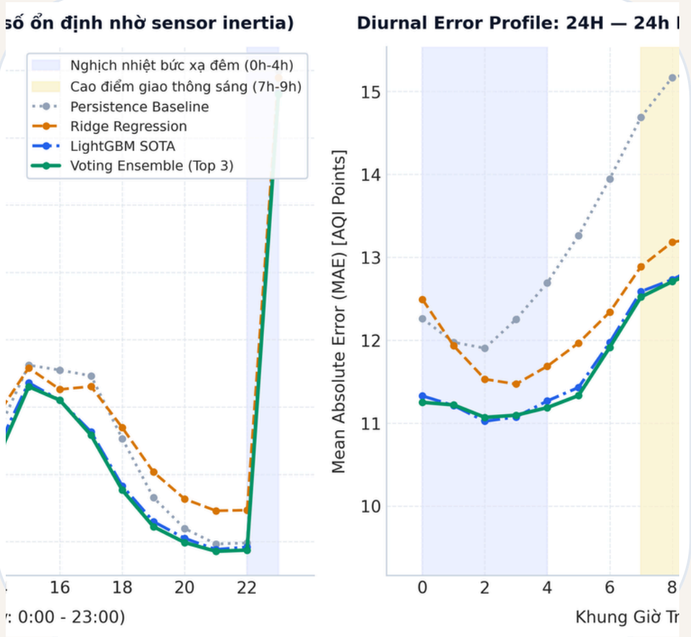


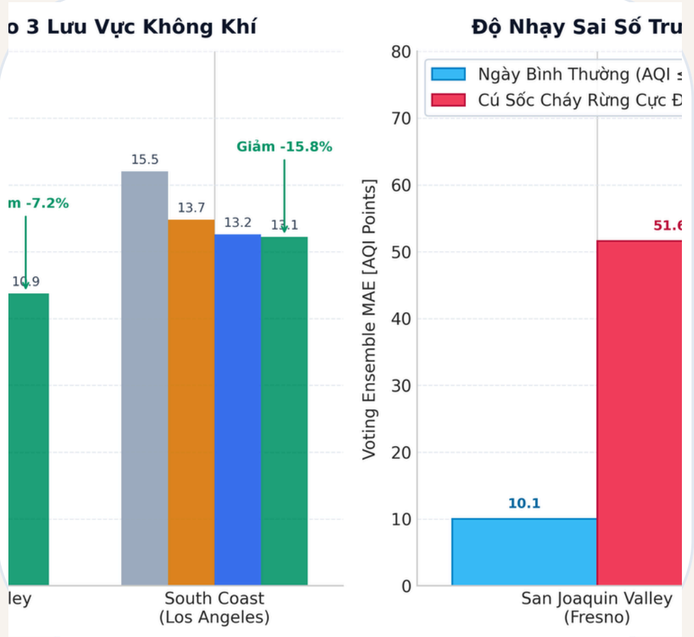
ERROR MECHANISMS & FORENSICS

Three-Dimensional Error Diagnostics & Topography

Empirical Diagnostics Across Tail Shrinkage, Nocturnal Inversion, and Basin Geomorphology







TAIL SHRINKAGE (β = 0.54)

Dense alignment on 1:1 line for safe levels (AQI <= 100). Regression slope β = 0.54 reveals symmetric MAE under-penalizes rare peaks, causing conservative tail shrinkage.

NOCTURNAL RADIATION INVERSION

MAE plunges at midday (~8 AQI), but surges at night & dawn (22h–4h >14 AQI) as radiative surface cooling creates a thermal lid, trapping particulate matter in a shallow boundary layer.

BASIN GEOMORPHOLOGY

San Jose MAE 9.90 (0 extremes, marine breeze) vs. Fresno MAE 10.95 (88 extremes, Central Valley bowl trap) vs. Los Angeles MAE 13.06 (231 extremes, San Gabriel barrier).

PART 4: CONCLUSION

Experimental Conclusions & Production Roadmap

Milestone 2 Wrap-up, Technical Limitations & Final Defense Deliverables

4 CORE EMPIRICAL FINDINGS

- Zero Leakage: Strict temporal split + trim buffer on 181k records
- Top Performer: Voting Ensemble leads 1h (MAE 3.06) & 24h (MAE 11.30)
- Efficiency: GBDT beats Deep MLP (+1.78 MAE) and trains 200x faster (0.41s)
- Diagnostics: Uncovered night inversion spikes & high-AQI shrinkage (β = 0.54)

3 TECHNICAL LIMITATIONS

- Peak Under-prediction: Symmetric MAE loss lags wildfire spikes (62.6 MAE)
- Vertical Blindspot: Ground sensors lack satellite smoke column data (AOD)
- Spatial Independence: Tabular models lack cross-basin spatial attention

MILESTONE 3 PRODUCTION ROADMAP

- Asymmetric Loss: Pinball / Huber loss to correct extreme slope (β → 1.0)
- Real-Time API: Containerized FastAPI serving predictions (<50ms latency)
- Early Warning Web App: Interactive dashboard for live California AQI alerts

Q&A DEFENSE SESSION

THANK YOU!